

Moolya

DevOps Implementation and Managed Operations

Enterprise-grade Automation, Observability, and Security at Scale

■ 14+

Years in Business

■ 200+

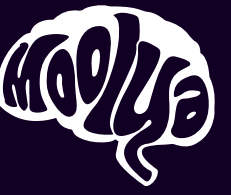
Customers

■ 20+

Countries

■ 80%

Industry leaders choose us



Who we are

At Moolya Software Testing Private Limited, we are a global leader in AI-driven engineering solutions, specializing in creating scalable, secure, and intelligent platforms that transform business outcomes. We collaborate with forward-thinking organizations to build high-impact products that leverage cutting-edge technologies such as Natural Language Processing (NLP), Generative AI, and advanced data analytics. Our expertise spans across AI platform engineering, software development, and quality assurance, ensuring measurable success for our clients.

Why Choose us



Skin in the Game

We go beyond vendor roles, fostering deep co-ownership with customers.



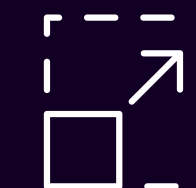
Culture Focus

We help customers address issues by focusing on cultural fixes.



Leadership Friendly

Moolya empowers C-, V-, and D-level leaders with insights.



Scaling Value

Moolya unifies Tech, Test, Product, Customer & Business insights to scale value.



Executive Summary

Keysight Technologies empowers innovators to design and deliver next-generation solutions with precision, speed, and confidence.

Moolya proposes a DevOps Implementation and Managed Operations solution to strengthen Keysight's software delivery ecosystem through a self-managed Kubernetes platform on AWS, integrated with your existing toolchain Jenkins, Polaris, Black Duck, and Vault.

This initiative will:

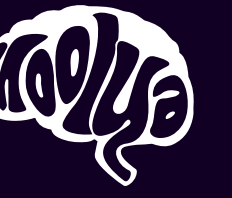
- Accelerate deployment cycles through automated CI/CD and secure pipelines.
- Enhance governance and compliance via Polaris and Black Duck integration.
- Ensure continuous reliability with Vault-based secrets, ACM certificates, and AWS observability.

Together, we'll enable Keysight to deliver faster, operate securely, and scale seamlessly, supporting your vision of engineering excellence.

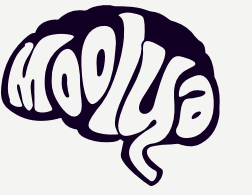


Engagement Objectives

- Drive **continuous improvement** and **governance** through measurable KPIs.
- Implement and manage **Kubernetes-based DevOps platform** on **AWS**.
- Integrate Jenkins pipelines with **Polaris** and **Black Duck** for compliance enforcement.
- Automate **certificate and secret lifecycle** via **Vault + AWS ACM**.
- Establish observability using **Prometheus, Grafana, CloudWatch**.
- Provide **executive visibility** through reports, dashboards, and audits.



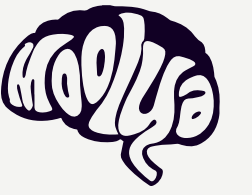
Current State vs. Moolya Value Delivered



From Maintenance Mode to Modern DevOps Excellence

Immediate operational continuity with parallel modernization — not a handoff, but a scale-up.

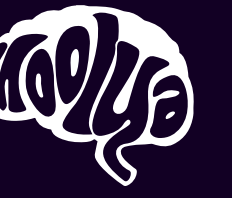
Area	Current	With Moolya Engagement	Value Delivered
Resourcing	1 Onshore DevOps engineer managing operations	3 Offshore DevOps engineers: architecture, automation, security	3× capability uplift with cross-domain expertise
Environment Management	Container operations managed manually on EC2/VMs	Kubernetes-based orchestration with automated scaling and rollback	Zero-downtime, predictable releases
CI/CD Process	--	Jenkins-based automated pipelines integrated with Polaris & Black Duck	60–70% faster build cycles; improved compliance
Security & Compliance	--	Integrated SAST/SCA gates, secrets rotation, certificate automation	Continuous compliance, reduced audit risk



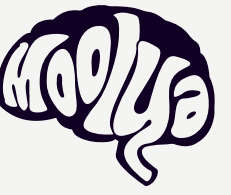
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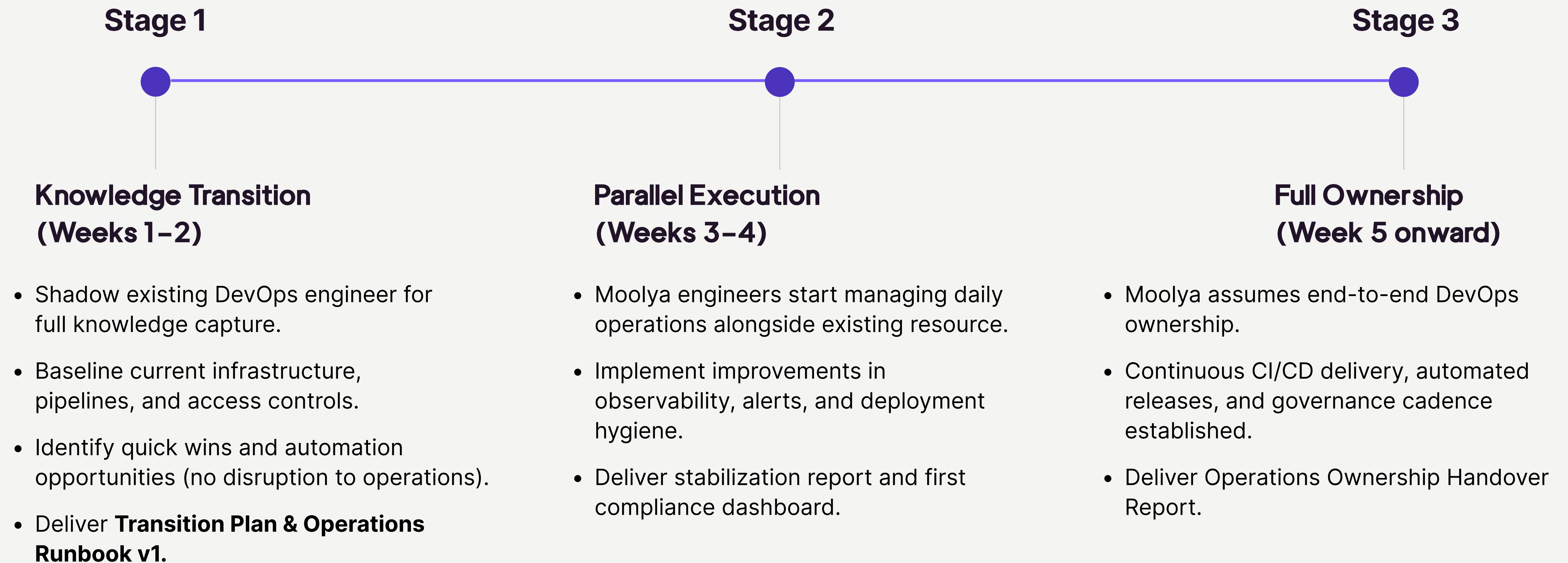
Area	Current	With Moolya Engagement	Value Delivered
Monitoring & Reporting	--	Prometheus + Grafana dashboards + SLA/SLO reporting	Real-time observability and proactive issue management
Operational Continuity	Dependency on a single resource	Dedicated Moolya team ensures 16×5 (US Timezone) continuity & governance	Risk-free transition and stable support
Governance & Optimization	--	Weekly + Monthly governance reviews with measurable KPIs	Transparent accountability and measurable ROI



Seamless Transition & Parallel Continuity Plan



Ensuring Zero Disruption During Transition





Implementing Kubernetes and CI/CD Enhancements Alongside Existing Workloads



Month 1 – Environment Design & Foundation Setup

Goal: Build AWS infrastructure, implement Kubernetes cluster, and establish the security foundation.

Week	Focus	Key Activities	Task Type
Week 1	Discovery & Assessment	Shadow existing DevOps engineer; assess current workload configurations while maintaining daily operations	One Time Task
Week 2	Architecture Design	Define AWS architecture (VPC, subnets, IAM), cluster node design, and RBAC model.	One Time Task
Week 3	Infrastructure Provisioning	Implement a self-managed Kubernetes cluster on AWS EC2 using Terraform, configuring networking, storage, and monitoring for operational readiness.	One Time Task
Week 4	Security & Certificates	Configure ACM, Vault, and namespace isolation; enforce Pod Security Standards and IAM policies.	Recurring Task

Deliverables:

- Implement Kubernetes clusters (Dev/Staging/Prod).
- Terraform IaC modules and state management repository.
- ACM-based SSL/TLS automation.
- Vault integration and initial secret policies.
- Security and access governance documentation.



Month 2 – Application Implementation & CI/CD Integration

Goal: Containerize workloads, automate CI/CD, and standardize secure deployment practices.

Week	Focus	Key Activities	Task Type
Week 5	Application Containerization	Refactor Dockerfiles, integrate Polaris (SAST) and Black Duck (SCA) scans in builds.	One Time Task
Week 6	Kubernetes Deployment	Deploy workloads to Kubernetes using Helm; configure Ingress and TLS with ACM.	One Time Task
Week 7	Jenkins Pipeline Setup	Implement CI/CD pipelines with build → scan → deploy → rollback flow; enforce Polaris & Black Duck gates.	Recurring Task
Week 8	Deployment Strategy Enablement	Implement Blue-Green and Canary deployments; integrate Prometheus-based health checks and rollback automation.	Recurring Task

Deliverables:

- Jenkins pipelines with security gates (Polaris, Black Duck).
- Helm-based deployment automation.
- Blue-Green and Canary rollout playbooks.
- Rollback and release validation runbooks
- CI/CD and compliance workflow documentation.



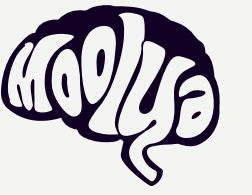
Month 3 – Observability, Validation & Operational Readiness

Goal: Final validation, observability setup, and knowledge transfer for operations.

Week	Focus	Key Activities	Task Type
Week 9	Observability Setup	Implement Prometheus, Grafana, and CloudWatch.	One Time Task
Week 10	Validation & Testing	Execute controlled Blue-Green and Canary deployments; validate SSL renewal and pipeline rollbacks.	One Time Task
Week 11	Compliance & Security Validation	Run full Polaris + Black Duck scans; verify secret rotation and certificate renewal.	Recurring Task
Week 12	Optimization & Knowledge Transfer	Optimize node sizing, finalize cost tuning, and conduct KT sessions with client team.	Recurring Task

Deliverables:

- Grafana dashboards and CloudWatch integrations
- Centralized Kubernetes log forwarding (CloudWatch)
- Final security & compliance validation reports
- Handover documentation with operational SOPs

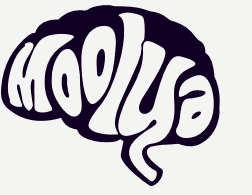


Certificate & Security Automation Framework

Layer	Tool	Description
Certificate Management	AWS ACM	Automated certificate issuance and renewal.
Secrets Management	Vault	Dynamic secret injection and rotation.
Ingress Management	NGINX	TLS termination, Blue-Green routing.
Static Code Analysis	Polaris	SAST integration in Jenkins pipeline.
Dependency Governance	Black Duck	SCA/BOM validation and license compliance.
Monitoring & Alerts	Prometheus + CloudWatch	Metrics and alerting for cert expiry & resource health.
Pipeline Enforcement	Jenkins	Build & deploy stage gating with audit trails.



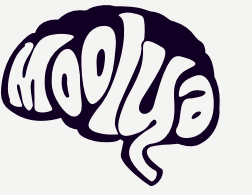
Evolving to a Fully Automated, Secure, and Always-On DevOps Ecosystem



1-Year Managed Operations & Optimization Plan

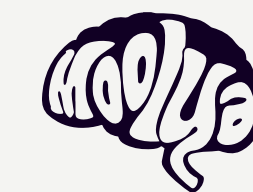
Quarterly Roadmap

Quarter	Objective	Refined Focus Areas (Clear, Outcome-Oriented)
Q1 (Months 1–3)	Establish and Operationalize Core DevOps Foundation	Implement Kubernetes environment on AWS EC2 using Terraform; implement CI/CD pipelines in Jenkins with integrated Polaris & Black Duck scans; enable Vault-based secrets and certificate automation.
Q2 (Months 4–6)	Stabilize Operations and Strengthen Compliance	Enhance observability through Prometheus, Grafana, and CloudWatch; automate compliance reporting; validate continuous certificate renewals and security posture.
Q3 (Months 7–9)	Optimize Performance and Scale Automation	Improve cost efficiency and resource utilization; enhance Jenkins pipeline performance; implement automated configuration drift detection and remediation.
Q4 (Months 10–12)	Institutionalize Governance and Drive Innovation	Establish DevOps governance KPIs and SLA dashboards; conduct resilience and reliability drills; introduce predictive observability and proactive alerting.



Ongoing Operations

Category	Activity	Frequency	Tool
Monitoring & Alerts	Infrastructure and cluster health monitoring	Continuous	Prometheus + CloudWatch
Code & Dependency Scanning	Jenkins integrated Polaris & Black Duck validation	Continuous	Jenkins
Secrets & Certificates	Validation and renewal	Continuous	Vault + ACM
Cost & Performance Optimization	Node pool right-sizing, resource scaling	Quarterly	AWS
Compliance Reporting	Security and compliance dashboards	Monthly	Grafana + Jenkins Reports
Disaster Recovery Simulation	Rollback, failover testing	Quarterly	Jenkins Automation
Governance Review	SLA/SLO health checks	Quarterly	Governance Dashboard



Technology Stack

Category	Tools
Cloud Infrastructure	AWS (EC2, VPC, IAM, ACM, Route 53)
Container Orchestration	Kubernetes (Self-managed)
Pipeline Management	Jenkins
Static Code Analysis	Polaris
BOM / Dependency Management	Black Duck
Secrets & Certificates	Vault + AWS ACM
IaC & Configuration	Terraform, Helm
Monitoring & Observability	Prometheus, Grafana, CloudWatch



Team Composition

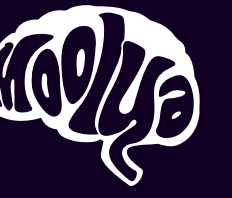
Role	Focus Area	Values Delivered
DevOps Lead	Solution design, infrastructure governance, automation strategy	Accelerates maturity; ensures long-term scalability
DevOps Engineer	Pipeline implementation, observability, deployment reliability	Delivers 70% faster releases; reduces incident volume
Security & Compliance Engineer	Polaris/Black Duck integration, vault management, audits	Continuous compliance; eliminates vulnerability backlogs

The combined team replaces the dependency on a single operator with an accountable, measurable, and outcome-driven function.

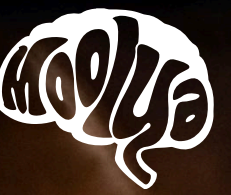


Expected Outcomes

- **Enterprise-grade Kubernetes environment** implemented on AWS.
- **Jenkins-based CI/CD pipelines** with integrated Polaris + Black Duck scans.
- **Certificate and secrets management** automated with ACM + Vault.
- **Unified observability stack** (Prometheus, Grafana, CloudWatch).
- **Executive governance dashboards** for SLA, cost, and compliance visibility.
- **Continuous optimization and compliance assurance** through managed operations.



Proven Case Studies



Case Study 1

Leading eCommerce Enterprise – DevOps Modernization and Continuous Delivery Excellence

About

- A leading global e-commerce platform serving millions of daily active users with peak-scale traffic during major shopping events.
- The engagement involved establishing a scalable, self-managed Kubernetes platform on AWS while maintaining full control of live workloads.

DevOps Value Delivered

- Took full DevOps ownership from Day 1 while implementing a self-managed Kubernetes platform on AWS EC2 using Terraform with zero downtime.
- Re-engineered Jenkins pipelines with automated build-test-deploy flow, integrated SonarQube for quality gates, and improved release reliability.
- Embedded OWASP Dependency-Check for open-source risk validation and baseline vulnerability scoring across all builds.
- Implemented HashiCorp Vault and AWS Certificate Manager (ACM) for dynamic secrets rotation and automated SSL renewals.

Business Impact

- SonarQube integration lowered high-severity issues by 70% and improved maintainability across key services.
- Terraform automation and AWS resource optimization reduced infrastructure costs by ~22%.
- OWASP Dependency-Check, Vault, and governance dashboards ensured continuous compliance and transparent reporting.



Case Study 2

Leading Supply Chain Management Platform

About the Platform

A supply chain and sustainability management platform designed for fashion brands across the European Union (EU), empowering them to manage operations and compliance efficiently using AWS.

DevOps Transformation

- Streamlined CI/CD Pipelines: Implemented end-to-end continuous integration and deployment pipelines using Cloudbees to ensure faster, automated, and error-free releases.
- Containerization & Infrastructure as Code (IaC): Migrated services to containerized environments and adopted IaC (Terraform + Kubernetes) for scalable and consistent infrastructure management.
- Automated Monitoring & Alerts: Set up real-time monitoring, logging, and alerting systems using Prometheus and Grafana to improve observability and system reliability.
- Environment Standardization: Unified staging, QA, and production environments to eliminate configuration drift and reduce deployment risks.

Business Impact

- Operational Agility: Enhanced platform reliability and enabled faster, more predictable releases.
- Cost Optimization: Reduced infrastructure overhead through automated scaling and resource utilization monitoring.



Case Study 3

Kubernetes-driven Delivery Transformation Using AWS

About the Platform

Enterprise SaaS client operating legacy EC2 deployments with manual release cycles and inconsistent configuration.

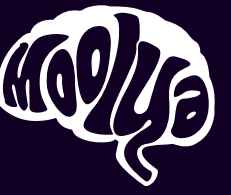
Moolya's Role:

- Designed AWS multi-account strategy with centralized IAM and GuardDuty.
- Containerized 40+ microservices and deployed to EKS clusters.
- Implemented Terraform modules for provisioning, networking, and IAM roles.
- Established Grafana dashboards for SLO tracking.

Results:

- Deployment frequency increased from bi-weekly to daily.
- Infrastructure drift reduced to near zero with IaC adoption.
- 35% cost optimization through right-sized autoscaling.
- Security exposure window reduced by 60% via automated scans.



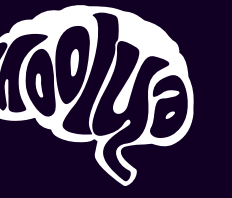


Conclusion

Moolya's DevOps Implementation and Managed Operations program establishes a secure, compliant, and fully automated DevOps ecosystem on AWS powered by Kubernetes, Jenkins, Polaris, and Black Duck.

Our deliverables provide not just automation, but visibility, governance, and confidence ensuring your infrastructure and pipelines remain continuously optimized, secure, and aligned with enterprise expectations.

This is not a one-time setup it's a sustained evolution of your DevOps maturity model, backed by transparent reporting and measurable results.



Thank You!